SSC MTS 2024 question paper with answers

29 October 2024 · Shift 2

82 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 8 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 16 questions

Q.3

Three cubes of sides 1 cm, 6 cm and 8 cm are melted to form a new cube. Find one-fourth of the surface area of the new cube.

(A) 101.5 cm^2
(B) 111.5 cm^2
(C) 131.5 cm^2
(D) 121.5 cm^2

Q.4

An item was sold for ₹40 with a profit of 15%. If it were sold for ₹36, then what would have been the percentage of profit or loss? (rounded off to 1 decimal places).

(A) Profit 4.7%
(B) Profit 3.5%
(C) Loss 4.5%
(D) Loss 3.7%

Q.5

The average of 7 numbers is 10.8. If the average of the first three be 10.5 and that of the last three is 11.4, find the value of the middle number.

(A) 9.7
(B) 9.8
(C) 9.9
(D) 9.5

Q.6

The lengths of two sides of a parallelogram are 6 cm and 20 cm. What is the sum of the squares of the diagonals of the parallelogram (in cm^2)?

(A) 728
(B) 800
(C) 782
(D) 872

Q.7

If 5 + x, 2x + 7, 6x + 9 and y are in proportion when x = 2, then find the value of y.

(A) 33
(B) 42
(C) 45
(D) 28

Q.8

The average of 11 consecutive numbers is 30. The smallest number among these is:

(A) 18
(B) 28
(C) 25
(D) 12

Q.10

Find the simple interest (in ₹) on ₹4000 as a sum borrowed at 4% per year rate of interest for 4 years.

(A) 600
(B) 640
(C) 580
(D) 680

Q.11

The average of 10 observations is 46. It was realised later that an observation was misread as 42 in place of 142. Find the correct average.

(A) 45
(B) 54
(C) 65
(D) 56

Q.12

On a circular race track of 2500 m, Sahil and Reena are running in the same direction at speeds of 27 km/h and 45 km/h, respectively. After how much time will they meet each other for the first time?

(A) 500 seconds
(B) 530 seconds
(C) 480 seconds
(D) 450 seconds

Q.14

Two trains of lengths 108 metre and 112 metre are running at speeds of 45 km/h and 54 km/h, respectively, towards each other on parallel tracks. In how many seconds will they cross each other?

(A) 8 seconds
(B) 9 seconds
(C) 12 seconds
(D) 10 seconds

Q.15

The average age of 6 children and their father is 13 years, and this average is reduced by 4 years if the father is excluded. How old is the father?

(A) 42 years
(B) 37 years
(C) 41 years
(D) 38 years

Q.16

If the radius of a right circular cylinder is decreased by 40% and its height is increased by 75%, its volume will be decreased by:

(A) 57%
(B) 37%
(C) 65%
(D) 35%

Q.17

The marked price of an item is 50% more than its cost price. The shopkeeper offers two successive discounts of 10% and 15%. What is his profit percentage?

(A) 15%
(B) 12.75%
(C) 15.75%
(D) 14.75%

Q.18

By selling an article for ₹20, a dealer makes a profit of 10%. By how much should he increase his selling price (in ₹, rounded off to two decimal place) so as to make a profit of 25%?

(A) 3.73
(B) 5.73
(C) 2.73
(D) 4.73

Q.19

In how many years will ₹4,000 invested at the rate of 18% per annum simple interest amounts to ₹5,800?

(A) 3.5
(B) 3
(C) 2
(D) 2.5

Q.20

Find the simple interest (in ₹) on ₹4000 as a sum borrowed at 4% per year rate of interest for 2 years.

(A) 360
(B) 280
(C) 320
(D) 260

Reasoning Ability and Problem Solving 16 questions

Q.1

TOVR is related to UPWS in a certain way based on the English alphabetical order. In the same way, QLSO is related to RMTP. To which of the following is NIPL related following the same logic?

(A) JOMQ
(B) JOQM
(C) OJMQ
(D) OJQM

Q.2

Read the given statement(s) and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statement(s). Statements: Some books are notes. All notes are copies. Conclusions: (I) Some books are copies. (II) No books is copies.

(A) Only conclusion (II) follows.
(B) Neither conclusion (I) nor (II) follows.
(C) Both conclusions (I) and (II) follow.
(D) Only conclusion (I) follows.

Q.4

S, U, B, J, E, C and T are sitting around a circular table facing the center (but not necessarily in the same order). C is sitting to the immediate right of T. U is an immediate neighbour of both T and E. Only one person is sitting between C and S when counted from the right of C. B is sitting to the immediate left of E. Who are the immediate neighbours of J?

(A) S and C
(B) U and B
(C) S and E
(D) B and E

Q.6

Select the letter-cluster from among the given options that can replace the question mark (?) in the following series. BCD, FGH, JKL, NOP, ?

(A) QRS
(B) QSR
(C) RST
(D) RQS

Q.7

In a certain code language, ‘rose are similar’ is written as ‘ki ko ku’ and ‘rose has thorns’ is coded as ‘ke ki ka’. How is ‘rose’ coded in that language?

(A) ku
(B) ko
(C) ka
(D) ki

Q.8

23 is related to 62 following a certain logic. Following the same logic, 37 is related to 104. To which ofthe following is 19 related, following the same logic?(NOTE: Operations should be performed on the whole numbers, without breaking down the numbersinto its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1and 3 is not allowed)

(A) 38
(B) 42
(C) 57
(D) 50

Q.9

P, Q, R, S, T and U are six friends sitting around a circular table, facing the centre (but not necessarily in the same order), having their breakfast. U is sitting exactly between Q and S and is third to the right of T. R is sitting to the immediate left of S. Who is the immediate neighbour of both P and R?

(A) U
(B) Q
(C) T
(D) S

Q.11

AS 9 is related to UU 2 in a certain way. In the same way, BT 6 is related to VV –1. To which of the following is HG 9 related, following the same logic?

(A) AJ 2
(B) AI 2
(C) BG –2
(D) BI 2

Q.12

What should come in place of the question mark (?) in the following series? 263, 268, 278, 293, 313, ?

(A) 341
(B) 340
(C) 339
(D) 338

Q.13

In a certain code language, 'ORANGE' is written as 'LIZMTV' and 'PURPLE' is written as 'KFIKOV'. What is the code for 'MAGENTA' in the given code language?

(A) NZTVMGZ
(B) NZTVNGZ
(C) NZSVMGZ
(D) NZTVMHZ

Q.14

In a certain code language, 'she can sing' is written as 'dq tk dp' and 'sing with me' is writtenas 'az dq js'. How is 'sing' writtenin the given language?

(A) tk
(B) dq
(C) dp
(D) az

Q.15

In the following number-pairs, the second number is obtained by applying certain mathematical operations to the first number. Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/ subtracting /multiplying to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) 34, 24; 85, 75

(A) 66, 56
(B) 36, 32
(C) 54, 52
(D) 42, 31

Q.16

What will come in place of the question mark (?) of the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 17 – 9 + 20 × 4 ÷ 5 = ?

(A) 26
(B) 1
(C) 20
(D) 15

Q.18

In a certain code language, 'written on paper' is written as 'ma rc gp' and 'paper and pen' is writtenas 'hk ma bo'. How will 'paper' be writtenin that language?

(A) gp
(B) ma
(C) hk
(D) bo

Q.19

What will come in the place of the question mark '?' in the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 39 + 3 ÷ 12 – 33 × 11 =?

(A) 5
(B) 4
(C) 3
(D) 6

Q.20

This question has two statements followed by two conclusions numbered I and II. You have to take the given statements to be true even if they seem to be at variance from commonly known facts. You have to decide which of the given conclusions logically follow(s) from the given statements. Statements: Some metres are litres. No litre is a gram. Conclusions (I) Some grams are not metres (II) All grams being metres is a possibility.

(A) Neither conclusion (I) nor (II) follows
(B) Only conclusion (I) follows
(C) Only conclusion (II) follows
(D) Both conclusions (I) and (II) follow

General Awareness 25 questions

Q.1

At which of the following places is the Lalbagh Flower Show organised, where thousands of flower varieties are put on display in different forms with a particular theme (which changes every year) during the exhibition?

(A) Hyderabad
(B) Patna
(C) Patiala
(D) Bengaluru

Q.2

Who among the following is the author of the autobiography 'Lone Fox Dancing: My Autobiography'?

(A) Khushwant Singh
(B) Brij Mohan Kaul
(C) John Dalvi
(D) Ruskin Bond

Q.3

Stunted growth, which is common in children growing up in extreme poverty, is a sure sign of _____.

(A) Undernutrition
(B) Sleep Deprivation
(C) Lack of Clothing
(D) Lack of Housing

Q.4

Which of the following is NOT a service of micro finance institution?

(A) Micro deposit
(B) Micro saving
(C) Micro insurance
(D) Micro loan

Q.5

The ‘Mangla’ oil field is located in which district of Rajasthan?

(A) Barmer
(B) Kota
(C) Bharatpur
(D) Jaipur

Q.6

When did Bindusara ascend the Maurya throne?

(A) 301 BCE
(B) 297 BCE
(C) 321 BCE
(D) 299 BCE

Q.7

According to Census of India 2011, which is the largest religious community in India?

(A) Sikhs
(B) Muslims
(C) Hindus
(D) Christians

Q.8

To which of the following classes of compounds does cyclohexane belong?

(A) Non-benzenoid aromatic compound
(B) Benzenoid aromatic compound
(C) Heterocyclic compound
(D) Alicyclic compound

Q.9

In which of the following years did the battle of Plassey take place?

(A) 1757
(B) 1787
(C) 1776
(D) 1796

Q.10

Women's Reservation Bill 2023 [The Constitution (One Hundred Twenty-Eighth Amendment) Bill, 2023] was introduced in Lok Sabha on _____.

(A) 20 Sep 2023
(B) 21 Sep 2023
(C) 19 Sep 2023
(D) 22 Sep 2023

Q.11

Which of the following Union Territories of India marks the lowest poverty in National Multidimensional Poverty Index: A Progress Review 2023?

(A) Chandigarh
(B) Lakshdweep
(C) Delhi
(D) Puducherry

Q.12

By what name is the species of Scaphopoda Mollusca 'Dentalium' commonly known?

(A) Duck mussel
(B) Tusk shells
(C) Heart clam
(D) Head foot

Q.13

Which Indian actress from the following, is also a popular dancer and received the Smita Patil Memorial award for her role in the movie Damini?

(A) Meenakshi Seshadri
(B) Jaya Prada
(C) Rekha
(D) Madhuri Dixit

Q.14

Which of the following Articles of the Constitution of India provides for the promotion of international peace and security?

(A) Article 47
(B) Article 44
(C) Article 49
(D) Article 51

Q.15

Which Hindustani classical musician is credited with popularising the Sarangi as a solo classical instrument?

(A) Pandit Shiv Kumar Sharma
(B) Pandit Ravi Shankar
(C) Pandit Ram Narayan
(D) Pandit Jnan Prakash Ghosh

Q.16

The Narora _____ Power Station is located in Uttar Pradesh.

(A) Natural Gas
(B) Wind
(C) Atomic
(D) Thermal

Q.17

Vedantam Satyanarayana Sarma is associated with which of the following dances?

(A) Kuchipudi
(B) Kathakali
(C) Odissi
(D) Mohiniyattam

Q.18

_____ was/were the mascot(s) of the Women's Asian Hockey Champions Trophy in 2023.

(A) Juhi
(B) Hansa
(C) Blaze and Tonk
(D) Yashi

Q.19

Groundwater is an important source of water. The top level of the groundwater layer is called:

(A) sea level
(B) underground water
(C) well
(D) water table

Q.20

Sanchi stupa was rediscovered in the year _____ by General Henry Taylor.

(A) 1817
(B) 1819
(C) 1818
(D) 1816

Q.21

If the server manages to land a legal serve and the receiver is unable to return the ball, it is known as an ace and the server gets the point. Which of the following sport is referred here?

(A) Wrestling
(B) Shooting
(C) Boxing
(D) Tennis

Q.22

Where of the following was Bal Gangadhar Tilak's Home Rule Movement active?

(A) Madras
(B) Bihar
(C) Punjab
(D) Maharashtra

Q.23

In which of the following cities was the headquarters of Theosophical Society established in British India?

(A) Adyar
(B) Ashok Nagar
(C) Alandur
(D) Ayanavaram

Q.24

Which of the following Parts of the Indian Constitution mentions about the relations between the Union and the States?

(A) Part VII
(B) Part XI
(C) Part X
(D) Part VIII

Q.25

In November 2023, who was appointed as the Chief Information Commissioner of India?

(A) Rajiv Gauba
(B) Heeralal Samariya
(C) Randeep Guleria
(D) Arvind Subramanian

English Language and Comprehension 25 questions

Q.1

Select the correctly spelled sentence.

(A) The liberian told the childrean to return their boooks by Friday.
(B) The librarian told the children to return their books by Friday.
(C) The libraian told the childrean to return their books by Friday.
(D) The librarian told the childern to return their books by Fryday.

Q.2

Select the INCORRECTLY spelt word.

(A) Eficcient
(B) Belligerent
(C) Tournament
(D) Parliament

Q.3

Select the most appropriate synonym of the underlined word in the following sentence. She is an “epitome” of grace.

(A) Example
(B) Admirer
(C) Destroyer
(D) Believer

Q.4

Change the following into superlative degree. She is taller than other girls in the class.

(A) In the class she is taller than other girls.
(B) She is the tallest girl in the class.
(C) No other girl in the class is as tall as she.
(D) She is tall girl in the class.

Q.5

Select the most appropriate ANTONYM of the word given in brackets to fill in the blank. The _____(sarcastic) comments by her husband made her feel uncomfortable, and she decided to walk out of the relationship.

(A) ruinous
(B) ambiguous
(C) harmonious
(D) gracious

Q.6

Select the most appropriate synonym of the underlined word. The speaker delivered an “engaging” speech that inspired the audience.

(A) captivating
(B) unnecessary
(C) dreary
(D) important

Q.7

Parts of the following sentence have been given as options. Select the option that contains an error. The 100-year-old banyan was blown off in the storm last night.

(A) in the storm
(B) was blown off
(C) last night
(D) The 100-year-old banyan

Q.8

Select the most appropriate option that can substitute the underlined segment in the given sentence. The teacher “called up the class” due to the sudden snowstorm.

(A) called on the class
(B) called out the class
(C) called off the class
(D) called in the class

Q.9

Select the appropriate option that can substitute the underlined word in the following sentence to make it grammatically correct. Meerut is “farthest” from Delhi.

(A) farther
(B) more farther
(C) far
(D) most farthest

Q.10
The following sentence has been split into four segments. Identify the segment that contains a grammatical error. The dog / eager wagged / its tail when / it saw its owner.

(A) The dog
(B) eager wagged
(C) it saw its owner
(D) its tail when

Q.11
The following sentence has been split into four segments. Identify the segment that contains a grammatical error. The soldiers / sacrifice their lives / to guarding / the nation and its boundaries.

(A) to guarding
(B) The soldiers
(C) the nation and its boundaries
(D) sacrifice their lives

Q.12
Select the most appropriate meaning of the given phrase. Reading between the lines

(A) Taking things literally
(B) Avoiding reading and engaging in gossips
(C) Reading a book
(D) Understanding a deeper or hidden meaning

Q.13
Identify the most appropriate ANTONYM of the word in brackets. The teacher commended the student for his (diligent) efforts.

(A) Conscientious
(B) Industrious
(C) Lazy
(D) Hardworking

Q.14
Select the most appropriate option to fill in the blank. She had a _____ for creating beautiful artwork.

(A) trigger
(B) talent
(C) post
(D) mode

Q.15
Select the most appropriate synonym of the given word. Meticulous

(A) Negligent
(B) Careless
(C) Precise
(D) Random

Q.16
Select the correct idiom to substitute the underlined segment in the following sentence. Roma was “the dearest child of her parents.”

(A) At sixes and sevens with her parents
(B) The apple of her parents’ eyes
(C) The star of her parents’ eyes
(D) The body and soul of her parents

Q.17
Select the option that can be used as a one-word substitute for the given phrase. One who is a newcomer

(A) Polyglot
(B) Nepotist
(C) Neophyte
(D) Narcissist

Q.18
The following sentence has been split into four segments. Identify the segment that contains a grammatical/spelling error. According to Mr. Sharma / the news have come / as another blow / to the family members.

(A) According to Mr. Sharma
(B) as another blow
(C) the news have come
(D) to the family members

Q.19

Select the most appropriate option to fill in the blank. The sudden plot twist in the story added an element of _____ that kept readers on the edge of their seats.

(A) monotony
(B) ambiguity
(C) predictability
(D) redundancy

Q.20

Select the INCORRECTLY spelt word. Majority of communication satelites are placed in geostationary orbits around the earth, which means that their speed matches the rate of its rotation.

(A) Satelites
(B) Communication
(C) Geostationary
(D) Rotation

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Despite her background role in the past, Ms Navalnaya has always been a crucial supporter of (1)..... husband and was instrumental in getting him out of Russia for (2)..... treatment when he was poisoned with Novichok nerve agent in 2020. She has been (3)..... as the 'First Lady' of the Russian opposition. And Alexei said himself that he could not continue his increasingly desperate and one-sided fight against the Kremlin without her. Their very public love story and family life - they have two children - (4)..... seen as an inspiration for their supporters. Yulia Ambrosimova was born in Moscow in 1976, daughter of the respected scientist Boris Ambrosimov. As economics graduate, she had a career in banking but (5)..... her job to bring up their two children when Alexei rose to prominence as an opposition politician.

Select the most appropriate option to fill in blank number 1.

(A) their
(B) them
(C) her
(D) hers

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Despite her background role in the past, Ms Navalnaya has always been a crucial supporter of (1)..... husband and was instrumental in getting him out of Russia for (2)..... treatment when he was poisoned with Novichok nerve agent in 2020. She has been (3)..... as the 'First Lady' of the Russian opposition. And Alexei said himself that he could not continue his increasingly desperate and one-sided fight against the Kremlin without her. Their very public love story and family life - they have two children - (4)..... seen as an inspiration for their supporters. Yulia Ambrosimova was born in Moscow in 1976, daughter of the respected scientist Boris Ambrosimov. As economics graduate, she had a career in banking but (5)..... her job to bring up their two children when Alexei rose to prominence as an opposition politician.

Select the most appropriate option to fill in blank number 2.

(A) interesting
(B) slow
(C) crazy
(D) urgent

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Despite her background role in the past, Ms Navalnaya has always been a crucial supporter of (1)..... husband and was instrumental in getting him out of Russia for (2)..... treatment when he was poisoned with Novichok nerve agent in 2020. She has been (3)..... as the 'First Lady' of the Russian opposition. And Alexei said himself that he could not continue his increasingly desperate and one-sided fight against the Kremlin without her. Their very public love story and family life - they have two children - (4)..... seen as an inspiration for their supporters. Yulia Ambrosimova was born in Moscow in 1976, daughter of the respected scientist Boris Ambrosimov. As economics graduate, she had a career in banking but (5)..... her job to bring up their two children when Alexei rose to prominence as an opposition politician.

Select the most appropriate option to fill in blank number 3.

(A) changed
(B) detected
(C) deleted
(D) described

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Despite her background role in the past, Ms Navalnaya has always been a crucial supporter of (1)..... husband and was instrumental in getting him out of Russia for (2)..... treatment when he was poisoned with Novichok nerve agent in 2020. She has been (3)..... as the 'First Lady' of the Russian opposition. And Alexei said himself that he could not continue his increasingly desperate and one-sided fight against the Kremlin without her. Their very public love story and family life - they have two children - (4)..... seen as an inspiration for their supporters. Yulia Ambrosimova was born in Moscow in 1976, daughter of the respected scientist Boris Ambrosimov. As economics graduate, she had a career in banking but (5)..... her job to bring up their two children when Alexei rose to prominence as an opposition politician.

Select the most appropriate option to fill in blank number 4.

(A) will been
(B) has been
(C) was being
(D) is been

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Despite her background role in the past, Ms Navalnaya has always been a crucial supporter of (1)..... husband and was instrumental in getting him out of Russia for (2)..... treatment when he was poisoned with Novichok nerve agent in 2020. She has been (3)..... as the 'First Lady' of the Russian opposition. And Alexei said himself that he could not continue his increasingly desperate and one-sided fight against the Kremlin without her. Their very public love story and family life - they have two children - (4)..... seen as an inspiration for their supporters. Yulia Ambrosimova was born in Moscow in 1976, daughter of the respected scientist Boris Ambrosimov. As economics graduate, she had a career in banking but (5)..... her job to bring up their two children when Alexei rose to prominence as an opposition politician.

Select the most appropriate option to fill in blank number 5.

(A) brought about
(B) gave up
(C) keep back
(D) broke down

Answer key

Numerical and Mathematical Ability

Q.3 **(D)**	Q.4 **(B)**	Q.5 **(C)**	Q.6 **(D)**	Q.7 **(A)**	Q.8 **(C)**
Q.10 **(B)**	Q.11 **(D)**	Q.12 **(A)**	Q.14 **(A)**	Q.15 **(B)**	Q.16 **(B)**
Q.17 **(D)**	Q.18 **(C)**	Q.19 **(D)**	Q.20 **(C)**		

Reasoning Ability and Problem Solving

Q.1 **(D)**	Q.2 **(D)**	Q.4 **(A)**	Q.6 **(C)**	Q.7 **(D)**	Q.8 **(D)**
Q.9 **(C)**	Q.11 **(D)**	Q.12 **(D)**	Q.13 **(A)**	Q.14 **(B)**	Q.15 **(A)**
Q.16 **(B)**	Q.18 **(B)**	Q.19 **(D)**	Q.20 **(C)**		

General Awareness

Q.1 **(D)**	Q.2 **(D)**	Q.3 **(A)**	Q.4 **(A)**	Q.5 **(A)**	Q.6 **(B)**
Q.7 **(C)**	Q.8 **(D)**	Q.9 **(A)**	Q.10 **(C)**	Q.11 **(D)**	Q.12 **(B)**
Q.13 **(A)**	Q.14 **(D)**	Q.15 **(C)**	Q.16 **(C)**	Q.17 **(A)**	Q.18 **(A)**
Q.19 **(D)**	Q.20 **(C)**	Q.21 **(D)**	Q.22 **(D)**	Q.23 **(A)**	Q.24 **(B)**
Q.25 **(B)**					

English Language and Comprehension

Q.1 **(B)**	Q.2 **(A)**	Q.3 **(A)**	Q.4 **(B)**	Q.5 **(D)**	Q.6 **(A)**
Q.7 **(B)**	Q.8 **(C)**	Q.9 **(C)**	Q.10 **(B)**	Q.11 **(A)**	Q.12 **(D)**
Q.13 **(C)**	Q.14 **(B)**	Q.15 **(C)**	Q.16 **(B)**	Q.17 **(C)**	Q.18 **(C)**
Q.19 **(B)**	Q.20 **(A)**	Q.21 **(C)**	Q.22 **(D)**	Q.23 **(D)**	Q.24 **(B)**
Q.25 **(B)**					

Explanations

Numerical and Mathematical Ability

Q.3 · Answer (D) 121.5 cm²

The new cube's volume is 1 + 216 + 512 = 729 cm³, so its side is 9 cm and its surface area 6 × 81 = 486 cm². One-fourth of that is 121.5 cm².

Q.4 · Answer (B) Profit 3.5%

The cost price is 40/1.15. Selling at ₹36 gives 36 × 1.15/40 = 41.4/40 = 1.035 times the cost: a profit of 3.5%.

Q.5 · Answer (C) 9.9

Total of all 7 = 7 × 10.8 = 75.6; first three = 31.5 and last three = 34.2. The middle number is 75.6 − 31.5 − 34.2 = 9.9.

Q.6 · Answer (D) 872

In a parallelogram the sum of the squares of the diagonals is $2(a^2 + b^2) = 2(6^2 + 20^2) = 2 \times 436 = 872$ cm².

Q.7 · Answer (A) 33

With x = 2 the terms are 7, 11, 21 and y. In proportion 7 : 11 = 21 : y, so y = 11 × 21/7 = 33.

Q.8 · Answer (C) 25

The average of 11 consecutive numbers is the middle (6th) one, 30, so the smallest is 30 − 5 = 25.

Q.10 · Answer (B) 640

SI = 4,000 × 4 × 4/100 = ₹640.

Q.11 · Answer (D) 56

The total was 10 × 46 = 460; correcting 42 to 142 adds 100, so the correct average is 560/10 = 56.

Q.12 · Answer (A) 500 seconds

Running the same way, Reena gains 45 − 27 = 18 km/h = 5 m/s on Sahil. They first meet when she has gained a full lap: 2500/5 = 500 seconds.

Q.14 · Answer (A) 8 seconds

Towards each other the relative speed is 45 + 54 = 99 km/h = 99 × 5/18 = 27.5 m/s. They cover 108 + 112 = 220 m in 220/27.5 = 8 seconds.

Q.15 · Answer (B) 37 years

All 7 together total 7 × 13 = 91 years; the 6 children alone average 9, a total of 54. The father is 91 − 54 = 37.

Q.16 · Answer (B) 37%

Volume varies as r^2h: $0.6^2 \times 1.75 = 0.36 \times 1.75 = 0.63$ of the original, a decrease of 37%.

Q.17 · Answer (D) 14.75%

Taking the cost as 100, the marked price is 150 and the selling price 150 × 0.9 × 0.85 = 114.75: a profit of 14.75%.

Q.18 · Answer (C) 2.73

The cost price is 20/1.1 = 200/11 ≈ ₹18.18. For a 25% profit he must sell at 1.25 × 200/11 = 250/11 ≈ ₹22.73, an increase of 250/11 − 20 = 30/11 ≈ ₹2.73.

Q.19 · Answer (D) 2.5

The interest is 5,800 − 4,000 = ₹1,800, and one year earns 4,000 × 18/100 = ₹720, so it takes 1,800/720 = 2.5 years.

Q.20 · Answer (C) 320

SI = 4,000 × 4 × 2/100 = ₹320.

Reasoning Ability and Problem Solving

Q.1 · Answer (D) OJQM
Each letter moves one place forward: TOVR → UPWS and QLSO → RMTP. So NIPL → OJQM.

Q.2 · Answer (D) Only conclusion (I) follows.
The books that are notes are also copies, so some books are copies (I). II contradicts I, so it cannot follow.

Q.4 · Answer (A) S and C
Going round to the right the order is B, E, U, T, C, J, S: C to T's immediate right, U between T and E, B to E's immediate left, and S second to the right of C, which leaves J between C and S. J's neighbours are S and C.

Q.6 · Answer (C) RST
Each cluster is three consecutive letters, and each starts 4 places after the last one began: B, F, J, N, R. The next is RST.

Q.7 · Answer (D) ki
'rose' is the only word in both phrases and 'ki' the only code in both, so rose = ki.

Q.8 · Answer (D) 50
The second number is three times the first minus 7: 23 × 3 − 7 = 62 and 37 × 3 − 7 = 104. Likewise 19 × 3 − 7 = 50.

Q.9 · Answer (C) T
With six seats, U third to the right of T sits opposite T. Q and S flank U, and R is to S's immediate left, so going round to the right from T the order is T, R, S, U, Q, P. P's neighbours are Q and T, and R's are T and S: T sits next to both.

Q.11 · Answer (D) BI 2
The first letter moves 6 back, the second 2 forward and the number falls by 7: AS 9 → UU 2 and BT 6 → VV −1. So HG 9 → BI 2.

Q.12 · Answer (D) 338
The differences are 5, 10, 15, 20, rising by 5 each time, so the next is 25: 313 + 25 = 338.

Q.13 · Answer (A) NZTVMGZ
Each letter is replaced by its mirror in the alphabet (A ↔ Z, B ↔ Y, ...): O → L, R → I, A → Z and so on. MAGENTA becomes NZTVMGZ.

Q.14 · Answer (B) dq
'sing' is the only word in both sentences and 'dq' the only code in both, so sing = dq.

Q.15 · Answer (A) 66, 56
In each pair the second number is 10 less than the first: 34 − 10 = 24 and 85 − 10 = 75. Likewise 66 − 10 = 56.

Q.16 · Answer (B) 1
After interchanging the signs: 17 + 9 − 20 ÷ 4 × 5 = 17 + 9 − 25 = 1.

Q.18 · Answer (B) ma
'paper' is the only word in both phrases and 'ma' the only code in both, so paper = ma.

Q.19 · Answer (D) 6
After interchanging the signs: 39 − 3 × 12 + 33 ÷ 11 = 39 − 36 + 3 = 6.

Q.20 · Answer (C) Only conclusion (II) follows
No litre is a gram, but the metres that are not litres may or may not be grams, so all grams being metres is possible (II follows) and 'some grams are not metres' is not certain (I does not follow).

General Awareness

Q.1 · Answer (D) Bengaluru
The Lalbagh Flower Show is held at the Lalbagh Botanical Garden in Bengaluru, around Independence Day and Republic Day.

Q.2 · Answer (D) Ruskin Bond
'Lone Fox Dancing: My Autobiography' (2017) is Ruskin Bond's account of his life.

Q.3 · Answer (A) Undernutrition
Stunting, being too short for one's age, is the mark of long-term undernutrition in childhood.

Q.4 · Answer (A) Micro deposit
SSC's key gives micro deposit. Microfinance is usually described as micro loans (microcredit), micro savings and micro insurance, and India's microfinance companies (NBFC-MFIs) may not accept deposits from the public; but savings are themselves a kind of deposit, so the line between the two options is thin.

Q.5 · Answer (A) Barmer
The Mangala oil field, discovered by Cairn in 2004, is in Barmer district of Rajasthan.

Q.6 · Answer (B) 297 BCE
Bindusara succeeded his father Chandragupta Maurya in about 297 BCE (dates for this period are approximate).

Q.7 · Answer (C) Hindus
In the 2011 Census Hindus made up about 79.8% of India's population, the largest religious community.

Q.8 · Answer (D) Alicyclic compound
Cyclohexane is a saturated ring of six carbon atoms with no aromatic character, so it is an alicyclic compound; benzene, by contrast, is benzenoid aromatic.

Q.9 · Answer (A) 1757
At Plassey, on 23 June 1757, the East India Company under Robert Clive defeated Siraj-ud-Daulah, the Nawab of Bengal.

Q.10 · Answer (C) 19 Sep 2023
The Constitution (One Hundred Twenty-Eighth Amendment) Bill, reserving a third of seats for women, was introduced in the Lok Sabha on 19 September 2023, the first day in the new Parliament building; it passed the Lok Sabha on 20 September and the Rajya Sabha on 21 September.

Q.11 · Answer (D) Puducherry
In NITI Aayog's 2023 review (NFHS-5 data, 2019-21) 0.85% of Puducherry's people were multidimensionally poor, against 1.11% in Lakshadweep, 3.43% in Delhi and 3.52% in Chandigarh.

Q.12 · Answer (B) Tusk shells
Dentalium belongs to the Scaphopoda, molluscs with tapering tubular shells open at both ends, commonly called tusk shells.

Q.13 · Answer (A) Meenakshi Seshadri
Meenakshi Seshadri, a trained classical dancer, played the title role in 'Damini' (1993), the year she received the Smita Patil Memorial Award.

Q.14 · Answer (D) Article 51
Article 51, a Directive Principle, asks the State to promote international peace and security, keep just and honourable relations between nations, respect international law and settle disputes by arbitration.

Q.15 · Answer (C) Pandit Ram Narayan
Pandit Ram Narayan made the sarangi, long an accompanying instrument, a solo concert instrument.

Q.16 · Answer (C) Atomic
The Narora Atomic Power Station is on the Ganga in Bulandshahr district of Uttar Pradesh.

Q.17 · Answer (A) Kuchipudi
Vedantam Satyanarayana Sarma was a Kuchipudi dancer, famous for female roles such as Satyabhama.

Q.18 · Answer (A) Juhi
Juhi, an elephant, was the mascot of the 2023 Women's Asian Champions Trophy, held in Ranchi and won by India.

Q.19 · Answer (D) water table
The water table is the upper surface of the saturated zone of groundwater.

Q.20 · Answer (C) 1818
General Henry Taylor, a British officer, was the first Westerner to record the Sanchi stupa, in 1818.

Q.21 · Answer (D) Tennis
In tennis an ace is a legal serve the receiver fails to touch, winning the point for the server.

Q.22 · Answer (D) Maharashtra
Tilak's Home Rule League, founded at Belgaum in April 1916, worked in Maharashtra (except Bombay city), Karnataka, the Central Provinces and Berar; Annie Besant's league covered the rest of India.

Q.23 · Answer (A) Adyar
The Theosophical Society moved its headquarters to Adyar, Madras (now Chennai), in 1882.

Q.24 · Answer (B) Part XI
Part XI (Articles 245 to 263) deals with the relations between the Union and the States.

Q.25 · Answer (B) Heeralal Samariya
Heeralal Samariya was sworn in as Chief Information Commissioner on 6 November 2023.

English Language and Comprehension

Q.1 · Answer (B) The librarian told the children to return their books by Friday.
Only this sentence spells 'librarian', 'children', 'books' and 'Friday' correctly.

Q.2 · Answer (A) Eficcient
The correct spelling is 'efficient'.

Q.3 · Answer (A) Example
An 'epitome' is a perfect example of a quality: 'example'.

Q.4 · Answer (B) She is the tallest girl in the class.
'Taller than other girls in the class' becomes, in the superlative degree, 'She is the tallest girl in the class'. 'No other girl ... as tall as she' is the positive degree.

Q.5 · Answer (D) gracious
'Sarcastic' remarks mock and wound; the opposite is 'gracious', kind and courteous.

Q.6 · Answer (A) captivating
'Engaging' means attractive and holding the attention: 'captivating'.

Q.7 · Answer (B) was blown off
A tree knocked over by the wind is 'blown down' (or uprooted); 'blown off' is said of something carried away from a surface, such as a roof or a hat.

Q.8 · Answer (C) called off the class
To 'call off' is to cancel, as a sudden snowstorm would make the teacher do; to 'call up' is to telephone or summon.

Q.9 · Answer (C) far
Meerut is not compared with any other place, so the positive degree is needed: 'Meerut is far from Delhi.'

Q.10 · Answer (B) eager wagged
A verb is modified by an adverb: 'The dog eagerly wagged its tail when it saw its owner.'

Q.11 · Answer (A) to guarding
The infinitive of purpose takes the base form of the verb: 'sacrifice their lives to guard the nation'.

Q.12 · Answer (D) Understanding a deeper or hidden meaning
To 'read between the lines' is to find a meaning that is implied rather than stated.

Q.13 · Answer (C) Lazy
'Diligent' means hard-working and careful; its opposite is 'lazy'. The other three are its synonyms.

Q.14 · Answer (B) talent
A 'talent' for something is a natural ability: 'a talent for creating beautiful artwork'.

Q.15 · Answer (C) Precise
'Meticulous' means showing great attention to detail: 'precise'. 'Negligent' and 'careless' are its opposites.

Q.16 · Answer (B) The apple of her parents' eyes
'The apple of someone's eye' is the person they cherish most.

Q.17 · Answer (C) Neophyte
A 'neophyte' is a newcomer or beginner. A polyglot speaks many languages, a nepotist favours relatives and a narcissist is excessively self-admiring.

Q.18 · Answer (C) the news have come
'News' is uncountable and takes a singular verb: 'the news has come'.

Q.19 · Answer (B) ambiguity
Only 'ambiguity', uncertainty about what will happen, could keep readers on the edge of their seats; monotony, predictability and redundancy would bore them.

Q.20 · Answer (A) Satelites
The correct spelling is 'satellites', with a double 'l'.

Q.21 · Answer (C) her
Ms Navalnaya supported her own husband, so the singular possessive: 'her husband'.

Q.22 · Answer (D) urgent
He had been poisoned with a nerve agent, so she got him out of Russia for 'urgent' treatment.

Q.23 · Answer (D) described
'She has been described as' the 'First Lady' of the Russian opposition: a label others have given her.

Q.24 · Answer (B) has been
The present perfect fits a view still held: their love story and family life, taken together as one idea, 'has been seen' as an inspiration. 'Will been' and 'is been' are not English, and the past continuous 'was being' does not fit.

Q.25 · Answer (B) gave up
She 'gave up', that is left, her banking job to bring up their two children.

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